Fiscal Year:	FY 2020	**Task Last Updated:**	FY 11/20/2020
PI Name:	Chang, Polly Y. Ph.D.		
Project Title:	Particle Radiation-Induced Dysregulation of Protein Homeostasis in the Brain		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	polly.chang@sri.com	**Fax:**	FY 650-859-2889
PI Organization Type:	NON-PROFIT	**Phone:**	650-859-2549
Organization Name:	SRI International		
PI Address 1:	Biosciences, PN 187		
PI Address 2:	333 Ravenswood Avenue		
PI Web Page:			
City:	Menlo Park	**State:**	CA
Zip Code:	94025-3453	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	05/01/2014	**End Date:**	08/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2022 per NSSC information (Ed., 2/6/22) NOTE: Project extended to 8/31/2021 with new grant 80NSSC20K1176, which has period of performance of 9/1/2020-8/31/2021 (Ed., 11/5/20) NOTE: Extended to 3/01/2021 per NSSC information (Ed., 2/27/2020) NOTE: Extended to 2/29/2020 per NSSC information (Ed., 10/16/19) NOTE: Extended to 10/31/2019 per NSSC information (Ed., 5/10/19) NOTE: Extended to 4/30/2019 per NSSC information (Ed., 4/11/18) NOTE: Change in period of performance (original period of performance was 2/26/2014-2/25/2018) per PI and NSSC information (Ed., 12/9/14)		
Key Personnel Changes/Previous PI:			

COI Name (Institution):	Lin, Hua Ph.D. (SRI International) Shaler, Thomas Ph.D. (SRI International)
Grant/Contract No.:	80NSSC20K1176 ; NNX14AE69G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The space atmosphere is comprised of wide spectrum of highly energetic and charged particle radiation. Exposure to this type of radiation in the galacto-cosmic environment or unplanned solar particle events (SPE) may pose a potential health hazard to humans in long-term manned space explorations. In normal biological systems, protein stability and system homeostasis are maintained through a tightly coupled post-translational modification process that is catalyzed by the ubiquitin-proteasome system for protein turnover. This process is necessary for modifying the correct protein in the correct cellular location at the correct time to control important processes, e.g., cell-cycle regulation, the maintenance of genomic stability, removal of damaged or misfolded proteins, autophagy, apoptosis, and gene transcription. Depending on the radiation dose and beam quality, particle radiation may cause significant damage to proteins and oxidative stress in the cells in the central nervous system, leading to dysfunction of the cellular ubiquitin-related pathways, an increase in the levels of extracellular ubiquitin, and persistent subsequent downstream effects on regulation of stress-associated signaling pathways in many different cell types within the central nervous system (CNS), precipitating neurodegenerative disease. We hypothesize that quantitative changes in cellular ubiquitin pools occur when neuronal cells are exposed to particle radiation and aim to study the impact of protein degradation after particle radiation using both in vitro and in vivo model systems. Information gathered from this research will aid in reducing the uncertainties associated with predicting risks both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate any potential negative effects in space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We expect that results associated with the regulation of protein processing and degradation in brain tissues obtained from our studies will be useful to understand molecular mechanisms of action in normal neuronal cell responses in particle radiation clinical therapy scenarios.
Task Progress:	Studies aimed at assessing the risk of neurodegeneration from exposure to the reactive oxygen species produced by HZE (high energy particle) radiation resulted in long-term production of oxidative stress. Among the molecular hallmarks of several neurodegenerative diseases, those manifesting from dysfunction of protein homeostasis in neurons resulting in the accumulation of ubiquitin-positive protein aggregates, are particularly prominent. The goal of our project is to further investigate radiation-quality dependent signaling mechanisms associated with protein homeostasis in brain tissues. Protein modification by ubiquitin is a highly regulated process catalyzed by enzyme families which can modify the correct protein in the correct cellular location at the correct time to control important processes such as cell cycle progression, removal of damaged or mis-folded proteins, and gene transcription. Currently, seven lysine residues have been identified as known points of ubiquitination, of which linkages at Lysine-11, -48, and -63 are the most abundant sites. Lysine 48-linked (designated as K-48) polyubiquitin chains target proteins for destruction by proteolysis. Lys-63-linked (also designated K-63) ubiquitination serves as a molecular platform for protein/protein interaction important for DNA damage repair, receptor endocytosis, kinase signaling activation, and protein trafficking. Utility of these markers is evidence by analytical results of gray matter in brain tissues from patients who died of Alzheimer disease where significantly increased K48, 11, and 63 linkages that correlated with increased Braak staging of disease severity. Justified by its emerging role in aging related CNS (central nervous system) diseases, the goal of our project is to dissect the unique role of these targeted protein degradation systems in particle radiation responses by quantitatively assess particle-radiation induced cellular protein turnover to uncover alterations in pathway-specific cellular ubiquitin pools and protein modification in vitro using cultured primary human and mouse neuronal stem cells, in vivo by evaluating particle radiation induced alterations in protein ubiquitination in brain tissues of C57Bl6 mice. In our studies, we measured levels of three major pUb linages using a number of cellular and animal models. Our data indicate that there are differences between how proton and iron particle beam irradiation affect pUb linkages that are associated with proteolysis (K48) and DNA damage repair protein recruitment (K63). The quality and quantity of DNA damage from protons and iron ions are well established in the literature. The fact that this effect is ‘sensed’ by the proteolytic process and has an impact at the post-translational modification level in neuronal cells is a unique and novel finding. Interestingly, our data from mouse embryonic brain cells showed regional specificity in that pUb changes were only observed in the striatal, but not in cortex and mesencephalon neurospheres. The relevance of our current work is supported by our early findings of protein ubiquitination in brain tissues in Alzheimer (AD) patients. The significance of brain ubiquitinome in degenerative brain pathology is further verified by a recent paper by Abreha et al. [Abreha et al. Proteomic, 18(20), 2018. PMID: 30230243] characterizing ubiquitin-positive pathological protein aggregates in 3 AD and 5 age-matched control human subjects. Using differential enrichment analysis, the authors in this paper showed that >800 ubiquitylation sites were significantly altered, representing an 80% increase in all of the 7 pUb linkages. Results from our work showed particle radiation quality plays an important role in the level, linkage, and spatial specificity of protein ubiquitination in neuronal cells and brain tissues. Our neuronal cell work has shown differential linkage utility depending of the LET (linear energy transfer) of the particle beam. At low LET, increased K48 pUb linkage is seen across the different cell types, while at high LET, increased K63 repair-associated linkages are seen in both human and mouse cells. Polyubiquitination in the hippocampus region of C57Bl6 mice was not significantly altered when compared to the controls for particles with LET =70 keV/µm. On the other hand, K48 linkages are elevated in a dose-dependent manner as early as 1 wk post irradiation for 1 GeV/u iron ions. Detailed analysis of hyperphosphorylation of microtubule associated protein Tau, core component of neurofibrillary tangles, showed that specific sites of Tau protein in specific brain regions (cerebellum and hippocampus) were hyperphosphorylated at 16 months after a low dose of iron ions. This significant finding points to the fact that particle radiation impacts the cross talk of ubiquitination and phosphorylation of Tau proteins and may have a long-term effect in the CNS.

Bibliography Type:	Description: (Last Updated: 04/26/2023)
Articles in Peer-reviewed Journals	Shaler T, Lin H, Bakke J, Chen S, Grover A, Chang P. "Particle radiation-induced dysregulation of protein homeostasis in primary human and mouse neuronal cells." Life Sci Space Res (Amst). 2020 May;25:9-17. Epub 2020 Feb 21. https://doi.org/10.1016/j.lssr.2020.02.003 ; PMID: 32414496 , May-2020